Retrospective benchmark of optical surface roughness workflows against an internal tactile baseline across multiple materials and processes

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Abstract

This paper presents a retrospective benchmark of optical surface-texture workflows against an internal tactile baseline across six engineering materials and multiple surface-generation processes. Eight profile roughness parameters were analysed. Optical–tactile discrepancy depended strongly on workflow choice. Under the best system–illumination combinations available in the archive, the lowest median discrepancies were obtained for amplitude parameters such as R_t, R_v, R_z, and R_{z1max}. Exported R_{sm} values required separate scale-sensitivity treatment: multiplying retained optical R_{sm} values by 1000 reduced the best-achievable median discrepancy from 99.9% to 17.3%, but 25% of material–process groups still remained above 30%. Across the strictly positive parameters, tactile repeatability was

generally smaller than the observed between-workflow discrepancies, with a median relative standard error of 1.45%. A fixed-configuration sensitivity analysis showed that the best single workflow with complete archive coverage had a median discrepancy of 24.8%, quantifying the effect of per-group retrospective selection. The resulting decision summaries separate lower-discrepancy candidate cases from high-discrepancy cases in which tactile confirmation remains preferred. The retrospective design, incomplete retention of optical repeat measurements and raw processing chains, and parameter-label comparison frame define the study as a workflow-level benchmark rather than as a formal optical–tactile equivalence assessment.
Keywords: surface metrology; optical measurement; tactile reference; surface roughness; illumination wavelength; benchmarking; uncertainty.

1 Introduction

Surface-texture measurement remains central to manufacturing, where reported roughness parameters directly influence process control, specification compliance, and product traceability. In industrial practice, acceptance criteria are still commonly anchored to profile parameters such as R_a, R_q, R_{sk}, R_{sm}, R_t, R_v, R_z, and R_{z1max}. These profile-parameter labels belong to the ISO geometrical product specification framework for profile surface texture and remain distinct from a full areal-parameter assessment under ISO surface-texture terminology [1,2]. Tactile stylus profilometry retains practical authority as a reference workflow, despite being slower, access-limited, and potentially intrusive on delicate or highly finished surfaces.

Recent work has improved both tactile and optical instrumentation, but in different ways. Imkamp et al. described navigator scanning as a route to higher-throughput tactile acquisition without sacrificing the scanning role of stylus metrology [3]. Linz et al. presented 3D fibre probes that combine tactile contact with optical read-out and very low probing force [4]. Mínguez-Martínez et al. reported interlaboratory comparison results that are directly relevant to profile-metrology comparability across implementations [5]. Boedecker et al. showed that white-light interferometer data require transfer-characteristic control and uncertainty-aware comparison before they can be meaningfully related to tactile results [6]. These developments expand the metrology toolbox while leaving a central comparability issue: optical and tactile systems may interrogate different effective measurands.

That comparability issue has been recognised repeatedly. Dierke and Tutsch showed that the material under test can itself bias optical results, particularly for transparent surfaces [7]. Tiziani et al. argued that microstructure measurement and description must account for material-specific optical behaviour [8]. García et al. compared contact stylus and confocal microscopy methods and showed that precision, accuracy, and algorithmic complexity depend on the chosen workflow [9]. Schmidt et al. compared areal and profile methods for machined components and highlighted that parameter validity can shift with application and material [10]. Nouira et al. built a high-precision profilometer precisely to compare tactile and optical measurements of standards under a controlled metrological frame [11]. Hoefling et al. then showed from an industrial perspective that large object size, reflective surfaces, contamination, and cycle-time demands can become hard method-selection constraints [12]. Stavroulakis et al. made the additional point that data fusion and algorithmic post-processing can materially change the final metrological output [13]. The practical question concerns the material–process conditions under which a given optical workflow remains acceptably close to an established tactile baseline.

Workflow behaviour may also vary with illumination and reconstruction strategy. Hu et al. developed a line-scanning chromatic confocal sensor specifically for highly reflective materials [14]. Hinz et al. compared fringe-projection systems across scale ranges and stressed application-specific sensor selection in production metrology [15]. Wang and Yang addressed saturation on highly reflective surfaces by combining adaptive and original-inverse fringe projection [16]. Schreiner et al. reduced speckle on optically rough technical surfaces by moving grazing-incidence interferometry into the infrared [17]. Tiziani et al. showed that spectral and temporal phase evaluation can extend interferometric analysis to rougher or stepped surfaces [18]. Sato and Ando recently proposed polynomial-exponent modelling of white-light scanning interferograms to improve envelope and phase identification [19].

Computational work places reconstruction and interpretation within the metrology chain. Xue et al. review machine-learning applications in optical metrology and show that phase demodulation, unwrapping, and phase-to-height conversion are now active modelling stages rather than passive post-processing [20]. Roberts et al. illustrate the same logic in industrial DUV–Vis–IR metrology, where machine learning is used to supplement conventional analysis models [21]. Chen et al. rely on spectrum-pattern matching to reconstruct chromatic confocal surface profiles [22]. Ekberg et al. use automated boundary detection to turn OCT data into dimensional measurements of multilayer ceramic structures [23]. Recent multi-task learning work also argues that shared models become attractive when multiple targets must be learned from limited data [24, 25]. Neethu and Vinod Chandra review how imbalance and unlabeled samples destabilize machine-learning performance [26], while Pouchard et al. document how platform and version differences can undermine reproducibility even when a model appears to transfer cleanly [27]. Heterogeneous retrospective datasets also constrain the level of inference supported by such approaches.

Against this background, the present work is framed as a retrospective benchmark rather than as an equivalence study. It analyses an archive spanning six engineering materials, multiple manufacturing processes, several optical workflow families, and multiple illumination bands. The study compares optical outputs with an internal tactile baseline for eight profile roughness parameters and quantifies where workflow discrepancy is low or high, how strongly that discrepancy depends on system, process, material, parameter, and illumination, and where tactile confirmation remains necessary.

The contribution is a structured benchmark of between-workflow discrepancy within the analysed archive. The metrological boundary is defined by the available comparison space: matched exported parameter labels are available, whereas a fully harmonised raw-profile measurand is not.

2 Methods

2.1 Study design and unit of analysis

This work is a retrospective benchmark based on an archived dataset of tactile and optical measurements acquired on six engineering materials and multiple surface-generation routes. Each observation is defined by specimen identity, roughness parameter, optical system, illumination condition, optical output, and the corresponding tactile result. The primary unit of comparison is the parameter–surface group, defined as one roughness parameter within one material–process combination. In the manuscript subset, each of the eight parameters contributes 59 material–process groups, yielding 472 parameter–surface groups in total. Not every optical workflow is represented in every group, so all analyses were performed on the available comparable subset for each task.

Archive specimen labels follow the pattern `<material>_<process>` (for example, `ti6al4v_mf` or `14301_wedm_f`). Material identifiers denote C45 steel (`c45`), AISI 304 stainless steel / EN 1.4301 (`14301`), Ti6Al4V (`ti6al4v`), Al7075 (`al7075`), MO58A brass (`mo58a`), and ELLOR graphite (`ellor`). Process identifiers denote rough milling (`mr`), finish milling (`mf`), rough and finish face turning (`tr`, `tf`), burnishing after finish milling (`b`), rough and finish wire electrical discharge machining (`wedm_r`, `wedm_f`), grinding (`gri`), glass bead blasting (`gla`), and honing (`hon`). The manuscript tables expand these archive codes for readability, whereas the exported archive files retain the short slugs.

The process labels denote archived surface-generation route classes rather than complete manufacturing recipes. Tool geometry, feed and speed settings, depths of cut, discharge settings for wire EDM, abrasive or honing conditions, and equivalent detailed process metadata were not available consistently across all materials. Material–process groups were analysed as benchmark strata rather than as levels in a designed manufacturing experiment.

2.2 Tactile baseline

Tactile stylus profilometry was treated as the internal baseline. It is the workflow in the archive for which repeated measurements were retained consistently across the manuscript subset. For each specimen and roughness parameter, the tactile mean

$$S = \frac{1}{n_S} \sum_{k=1}^{n_S} T_k$$

was computed from $n_S = 20$ repeats, with accompanying standard deviation s_S and standard error $u_S = s_S/\sqrt{n_S}$. These quantities represent internal repeatability and are reported separately from traceability-qualified uncertainty.

2.3 Optical workflows and comparison space

The optical archive includes confocal microscopy, focus variation, confocal–interferometric fusion, and interferometry-based systems, with multiple illumination bands where available. System abbreviations used in the figures and tables follow the archive nomenclature: *conf* for confocal microscopy, *FV* for focus variation, *fusion* for confocal–interferometric fusion, and *int* for interferometry/coherence-scanning interferometry.

Comparison was restricted to matched specimen identity and same-named exported roughness parameters. The retained optical metadata identify workflow family and illumination condition; model-specific objectives, numerical apertures, lateral sampling, vertical resolution, filtering, form removal, and reconstruction settings are incomplete across the archive. A common raw-profile measurand was outside the retained archive fields, which lack a complete common processing chain across instruments, including sampling, filtering, form removal, lateral resampling, and signal-processing settings. The reported discrepancies are between-workflow discrepancies within this exported-parameter comparison space.

2.4 Response variables

The parameter symbols follow the profile roughness notation used in ISO surface-texture terminology. The comparison is restricted to same-named exported profile parameters; areal surface-texture parameters were outside the retained manuscript subset. For the seven strictly positive roughness parameters (R_a, R_q, R_{sm}, R_t, R_v, R_z, and R_{z1max}), discrepancy between optical output O and tactile baseline S was defined as

$$d_p = 100 \left| \frac{O - S}{S} \right| . \tag{1}$$

For the sign-sensitive parameter R_{sk}, near-zero tactile denominators make percentage normalisation unstable. Archive-derived decision or regression summaries that retain percentage-based R_{sk} values are treated as secondary screening descriptors, separately from the strictly positive parameters.

2.5 Decision-oriented summaries

Within each parameter–surface group, optical workflows were ranked by the available observed discrepancy, using the median when more than one retained optical output was available for the same system–illumination pair. In the archived decision subset, optical replication is incomplete; most system–illumination entries represent one exported optical result matched to a tactile mean. The decision tables report the minimum retained discrepancy across the tested system–illumination combinations and identify the workflow that achieved it. These retrospective summaries describe best-achievable behaviour within the archive, separately from the performance expected for a single fixed optical configuration.

To quantify this optimisation effect, a fixed-configuration sensitivity analysis was performed for the seven strictly positive parameters. Each candidate optical workflow was defined as one system–illumination pair and was evaluated across all parameter–surface groups in which that pair was available, without allowing the best workflow to vary by group. The resulting fixed-workflow medians were compared with the retrospective best-achievable medians from the same groups. Coverage was reported relative to the 413 strictly positive parameter–surface groups used in the decision summaries.

The reproducibility record lists the available analysis fields: material and process identifiers, specimen-level parameter labels, optical system, illumination condition, optical roughness outputs, tactile repeat measurements, and derived discrepancy summaries. Raw profiles, a shared filtering/form-removal chain, full instrument settings, and a combined tactile–optical uncertainty budget are incomplete across the archive. Accordingly, the release supports a retrospective workflow benchmark rather than a prospective traceability-oriented validation study.

2.6 Statistical analysis

Statistical analyses were performed within parameter–surface groups. Two-factor ANOVA with factors *system* and *illumination colour* provided an exploratory decomposition of variance. In this retrospective, unbalanced archive, nominal p-values served as screening statistics; interpretation was based primarily on effect sizes and on consistency of patterns across groups.

System-wise linear regressions of discrepancy versus nominal illumination wavelength were used as low-degree-of-freedom descriptive summaries. Most fits rely on three or four wavelength levels and were treated as screening trends rather than physical calibration laws.

Non-parametric percentile bootstrap intervals were computed for the aggregate median best-achievable discrepancies by parameter, material, and process. Bootstrap resampling was performed across archive-defined parameter–surface groups within each aggregate stratum, using 2000 resamples and a fixed random seed; these intervals describe stability of the retrospective median summaries and are not prospective validation intervals.

Supplementary tables and representative figures were generated from the same exported analysis outputs used in the main manuscript. The supplement generator (`generate_supplement.py`) formats the ANOVA screens, wavelength-regression summaries, system descriptives, tactile repeatability summaries, bootstrap stability summaries, and selected representative figures without adding a separate statistical model.

To compare workflow similarity within a given parameter–surface group, a technical distance matrix was defined from mean discrepancies. If $\bar{d}_i$ and $\bar{d}_j$ denote mean discrepancy for techniques i and j, where a technique is one optical system under one illumination condition, then

$$D_{ij} = |\bar{d}_i - \bar{d}_j|. \tag{2}$$

This quantity describes similarity of disagreement patterns within the analysed dataset, separately from traceability, matched calibration, or measurand equivalence.

3 Results

Across the manuscript subset, each of the eight parameters contributes 59 material–process groups, yielding 472 parameter–surface groups overall, all with tactile baselines based on $n_S = 20$ repeats. For the seven strictly positive parameters, the relative standard error of the tactile mean had median 1.45% and interquartile range 0.85–2.44%; parameter-wise summaries are reported in Supplementary Table S4. Tactile repeatability is generally smaller than the between-workflow discrepancies analysed below, supporting the use of the tactile mean as an internal benchmark baseline.

3.1 Decision-oriented benchmark summaries

For each parameter–surface group, the decision summaries retain the lowest observed median disagreement across the tested system and illumination combinations. These tables describe best-achievable performance within the archive, separately from the behaviour of any single fixed configuration.

Table 1 summarises best-achievable median discrepancy for the seven strictly positive parameters. Among them, amplitude-type measures (R_t, R_v, R_z, and R_{z1max}) achieve the lowest medians, ranging from 4.8% to 6.6%; R_a and R_q remain below 10% in median terms, whereas R_{sm} remains problematic at 99.9%. No single optical workflow dominates all parameters, although interferometry-based configurations are frequently among the best-performing options. Percentage-based R_{sk} rankings are excluded from the manuscript decision tables due to near-zero tactile denominator instability; they are retained as archive-level screening outputs.

The band columns in Table 1 provide a practical triage view rather than a normative acceptance rule. Groups in the lowest band identify candidate optical workflows for prospective protocol validation, groups in the middle band indicate cases where material–process-specific checking remains important, and groups above 30% describe conditions where tactile confirmation or a revised optical protocol remains the more defensible route. The same interpretation separates R_{sm} from the amplitude parameters: all 59 R_{sm} groups remain in the high-discrepancy band, whereas the lower-discrepancy band contains 63–69% of the R_t, R_v, R_z, and R_{z1max} groups.

Table 1: Decision-oriented summary of optical–tactile agreement by strictly positive roughness parameter (best system+colour chosen per material–process group)

Parameter	N	Median best |diff| [%]	Q1–Q3	$\leq$ 10%	10–30%	>30%	Most frequent best (system, colour)
R_a	59	9.8	2.6–21.8	30 (51%)	16 (27%)	13 (22%)	int, red
R_q	59	7.4	2.4–22.5	35 (59%)	12 (20%)	12 (20%)	int, red
R_{sm}	59	99.9	99.9–99.9	0 (0%)	0 (0%)	59 (100%)	int, white
R_t	59	6.6	2.1–13.2	39 (66%)	13 (22%)	7 (12%)	int, green
R_v	59	4.8	1.3–19.0	40 (68%)	7 (12%)	12 (20%)	int, red
R_z	59	5.3	1.2–20.3	37 (63%)	10 (17%)	12 (20%)	int, red
R_{z1max}	59	5.7	2.0–12.3	41 (69%)	10 (17%)	8 (14%)	int, red

Note: N is the number of available material–process groups for the seven strictly positive parameters. For each group, the optical configuration (system+illumination colour) that minimises the median $|\text{diff}|(\%)$ is selected. Q1–Q3 are quartiles of the selected medians. The band columns ($\leq$ 10%, 10–30%, >30%) count how many groups fall into each range; values are shown as count (percent of N), e.g. 7 (15%) means 7 groups, i.e. 15% of N. Most frequent best denotes the modal selected (system, colour) across groups. Percentage-based R_{sk} rankings are excluded from the manuscript decision tables because normalisation becomes unstable near zero; archive-level screening rows are retained separately in the exported outputs.

Because R_{sm} was the only strictly positive parameter that remained uniformly in the high-discrepancy band, it was segmented separately by optical system, material, and process (Table 2). The diagnostic segmentation shows that the effect is not concentrated in one workflow family or one surface class. For all four optical-system families, every retained R_{sm} system–colour entry exceeded 30% discrepancy. After retrospective best-workflow selection, every material stratum and every process stratum also remained entirely above 30% discrepancy.

Table 2: Diagnostic segmentation of R_{sm} optical–tactile discrepancy by optical system, material, and process

View	Stratum	N	Median |diff| [%]	Q1–Q3	$\leq 10\%$	>30%	Modal workflow
Optical system	conf	172	100.0	99.9–100.0	0 (0%)	172 (100%)	all colours
Optical system	FV	177	99.9	99.9–99.9	0 (0%)	177 (100%)	all colours
Optical system	fusion	177	100.0	100.0–100.0	0 (0%)	177 (100%)	all colours
Optical system	int	233	99.9	99.9–99.9	0 (0%)	233 (100%)	all colours
Material	AISI 304 / EN 1.4301 (14301)	10	99.9	99.9–99.9	0 (0%)	10 (100%)	FV, white
Material	Al7075	10	99.9	99.8–99.9	0 (0%)	10 (100%)	FV, white
Material	C45 steel	10	99.9	99.9–99.9	0 (0%)	10 (100%)	int, white
Material	Graphite (ELLOR)	9	99.9	99.9–99.9	0 (0%)	9 (100%)	int, blue
Material	MO58A brass	10	99.9	99.9–99.9	0 (0%)	10 (100%)	FV, green
Material	Ti6Al4V	10	99.9	99.9–99.9	0 (0%)	10 (100%)	int, white
Process	MR (rough milling)	6	99.9	99.8–99.9	0 (0%)	6 (100%)	int, white
Process	MF (finish milling)	6	99.9	99.9–100.0	0 (0%)	6 (100%)	int, white
Process	TR (rough face turning)	6	99.8	99.7–99.8	0 (0%)	6 (100%)	int, white
Process	TF (finish face turning)	6	99.8	99.6–99.9	0 (0%)	6 (100%)	int, white
Process	B (burnishing after MF)	5	99.9	99.9–100.0	0 (0%)	5 (100%)	int, white
Process	WEDM_R (rough wire EDM)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	int, blue
Process	WEDM_F (finish wire EDM)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green
Process	GRI (grinding)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green
Process	GLA (glass bead blasting)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green
Process	HON (honing)	6	99.9	99.9–99.9	0 (0%)	6 (100%)	FV, green

Note: Optical-system rows summarise all retained R_{sm} system–colour entries. Material and process rows summarise the retrospective best-achievable R_{sm} discrepancy per material–process group. The modal workflow is the most frequently selected best system+colour within the corresponding material or process stratum; for optical-system rows, all retained illumination colours are pooled.

The retained R_{sm} comparison was then examined for unit-scale sensitivity because optical spacing exports were numerically much smaller than the tactile spacing values. Multiplying optical R_{sm} values by 1000, treated as a post hoc millimetre-to-micrometre sensitivity rather than as a primary correction, reduced the all-entry median discrepancy from 99.9% to 47.4% and the best-workflow median across material–process groups from 99.9% to 17.3% (Table 3). Under this sensitivity, 24 of 59 best-workflow groups moved into the $\leq 10\%$ band, while 15 of 59 remained above 30%.

Table 3: Sensitivity of R_{sm} discrepancy to retained optical unit scale

Comparison set	Scale assumption	N	Median |diff| [%]	Q1–Q3	$\leq$ 10%	>30%	Median tactile/optical ratio
All retained system-colour entries	Exported optical value	761	99.9	99.9–100.0	0 (0%)	761 (100%)	1799.8
All retained system-colour entries	Optical value multiplied by 1000	761	47.4	28.0–66.0	53 (7%)	558 (73%)	1.8
Best workflow per material-process group	Exported optical value	59	99.9	99.9–99.9	0 (0%)	59 (100%)	1030.4
Best workflow per material-process group	Optical value multiplied by 1000	59	17.3	3.9–30.2	24 (41%)	15 (25%)	1.2

Note: The x1000 rows are a sensitivity check that treats retained optical R_{sm} values as if they were exported in millimetres and converted to micrometres before comparison. They are not used as corrected primary endpoints because complete original unit metadata and raw processing chains are not retained consistently across the archive.

Tables 4 and 5 aggregate the same best-achievable summaries by surface-generation process and by material across the 413 strictly positive parameter–surface groups. Even after allowing the best system and illumination to vary, process and material still strongly modulate how often optical workflows enter a low-disagreement regime.

Bootstrap percentile intervals for the aggregate medians are reported in Supplementary Table S5. They support the same hierarchy as the point estimates: amplitude-parameter medians remain in the lower-discrepancy range, retained exported-value R_{sm} remains close to 100%, and process-level intervals are wider where the archive combines more heterogeneous surface-generation routes.

Table 4: Decision summary aggregated by surface-generation process across all materials and strictly positive parameters

Process	N	Median best |diff| [%]	Q1–Q3	$\leq$ 10%	10–30%	>30%	Most frequent best (system, colour)
GLA (glass bead blasting)	42	4.7	1.1–7.5	35 (83%)	1 (2%)	6 (14%)	FV, blue
WEDM_R (rough wire EDM)	42	1.6	0.6–8.1	33 (79%)	3 (7%)	6 (14%)	int, green
HON (honing)	42	3.8	1.2–12.1	30 (71%)	5 (12%)	7 (17%)	int, red
WEDM_F (finish wire EDM)	42	7.5	2.9–12.9	29 (69%)	7 (17%)	6 (14%)	FV, green
MF (finish milling)	42	8.1	2.0–24.0	23 (55%)	11 (26%)	8 (19%)	int, green
GRI (grinding)	42	11.2	2.8–94.7	18 (43%)	8 (19%)	16 (38%)	int, blue
MR (rough milling)	42	28.6	4.7–42.7	17 (40%)	4 (10%)	21 (50%)	int, blue
TF (finish face turning)	42	25.3	3.0–50.2	16 (38%)	5 (12%)	21 (50%)	fusion, red
TR (rough face turning)	42	31.2	7.6–75.0	12 (29%)	9 (21%)	21 (50%)	fusion, red
B (burnishing after MF)	35	19.0	9.7–37.7	9 (26%)	15 (43%)	11 (31%)	int, white

Note: N is the number of contributing strictly positive parameter–material–process groups. For each group, the optical configuration (system+illumination colour) that minimises the median $|\text{diff}|(\%)$ is selected. Q1–Q3 are quartiles of the selected medians. The band columns ($\leq$ 10%, 10–30%, >30%) count how many groups fall into each range; values are shown as count (percent of N), e.g. 7 (15%) means 7 groups, i.e. 15% of N. Most frequent best denotes the modal selected (system, colour) across groups. Archive identifiers in the first column are expanded for readability.

Table 5: Decision summary aggregated by material across all processes and strictly positive parameters

Material	N	Median best |diff| [%]	Q1–Q3	$\leq$ 10%	10–30%	>30%	Most frequent best (system, colour)
Graphite (ELLOR)	63	4.6	0.9–19.8	42 (67%)	6 (10%)	15 (24%)	int, green
C45 steel	70	7.3	1.2–54.6	40 (57%)	6 (9%)	24 (34%)	int, red
MO58A brass	70	8.2	3.4–33.1	38 (54%)	13 (19%)	19 (27%)	int, blue
Al7075	70	8.2	3.0–41.0	37 (53%)	11 (16%)	22 (31%)	int, red
AISI 304 / EN 1.4301 (14301)	70	10.9	1.6–33.3	33 (47%)	17 (24%)	20 (29%)	FV, green
Ti6Al4V	70	11.7	4.4–51.6	32 (46%)	15 (21%)	23 (33%)	int, red

Note: N is the number of contributing strictly positive parameter–material–process groups. For each group, the optical configuration (system+illumination colour) that minimises the median $|\text{diff}|(\%)$ is selected. Q1–Q3 are quartiles of the selected medians. The band columns ($\leq$ 10%, 10–30%, >30%) count how many groups fall into each range; values are shown as count (percent of N), e.g. 7 (15%) means 7 groups, i.e. 15% of N. Most frequent best denotes the modal selected (system, colour) across groups. Archive identifiers in the first column are expanded for readability.

The fixed-configuration sensitivity analysis makes the retrospective-selection effect explicit (Table 6). The best single workflow with complete coverage was interferometry with green illumination, with median discrepancy 24.8% across all 413 strictly positive parameter–surface groups and $\leq 10\%$ discrepancy in 93 groups (23%). Interferometry with white illumination achieved a similar median discrepancy (25.0%) but with slightly lower coverage (406/413 groups). These fixed-workflow medians are substantially higher than the best-achievable medians in Table 1, showing that the low discrepancies in the decision summaries depend on parameter- and surface-specific workflow selection.

Table 6: Fixed-configuration sensitivity analysis for the strictly positive roughness parameters

Fixed workflow	N	Coverage	Median $\lvert$diff$\rvert$ [%]	Q1–Q3	$\leq 10\%$	>30%	Median excess vs. best [%]
int, green	413	100%	24.8	11.1–66.9	93 (23%)	198 (48%)	7.2
int, white	406	98%	25.0	10.8–66.7	92 (23%)	189 (47%)	8.0
int, red	413	100%	31.5	12.1–90.5	87 (21%)	212 (51%)	9.1
FV, green	413	100%	35.4	14.9–86.7	67 (16%)	232 (56%)	13.7
FV, blue	413	100%	40.2	15.5–99.9	66 (16%)	235 (57%)	11.5
int, blue	399	97%	43.0	19.9–90.3	48 (12%)	249 (62%)	13.5
conf, blue	378	92%	46.5	18.2–99.9	62 (16%)	236 (62%)	17.8
FV, red	413	100%	47.2	19.0–99.7	69 (17%)	265 (64%)	17.5

Note: Fixed workflow means that one optical system+illumination pair is evaluated wherever it is available, without selecting a different best configuration for each parameter–surface group. Coverage is relative to the 413 strictly positive parameter–surface groups used in the manuscript decision summaries. The table shows the eight lowest median fixed workflows; the complete fixed-workflow ranking and group-level medians are exported as CSV files. Excess vs. best is the median difference between the fixed-workflow median discrepancy and the retrospective best-achievable median discrepancy within the same group.

Read together, Tables 1 and 6 separate two operational decisions. The best-achievable summaries describe where the archive contains an optical workflow worth validating for a specific parameter–surface group. The fixed-workflow summaries describe the consequence of choosing one optical setup before measurement across a broad material–process range. This distinction is central to the benchmark: the archive contains many lower-discrepancy local matches, but the best complete-coverage fixed workflow still leaves nearly half of the strictly positive parameter–surface groups above 30% discrepancy.

Figure 1 summarises the same best-achievable benchmark as a process–parameter map. It is derived from the same retrospective best-workflow selection logic as Table 1, but retains the process dimension. It highlights two practically important patterns. First, the retained exported-value R_{sm} comparison remains close to 100% median discrepancy across all processes even after retrospective workflow selection. Second, amplitude parameters show strong process dependence: rough wire EDM, honing, glass bead blasting, and finish wire EDM generally occupy the low-discrepancy region, whereas rough face turning and rough milling contain several higher-discrepancy cells.

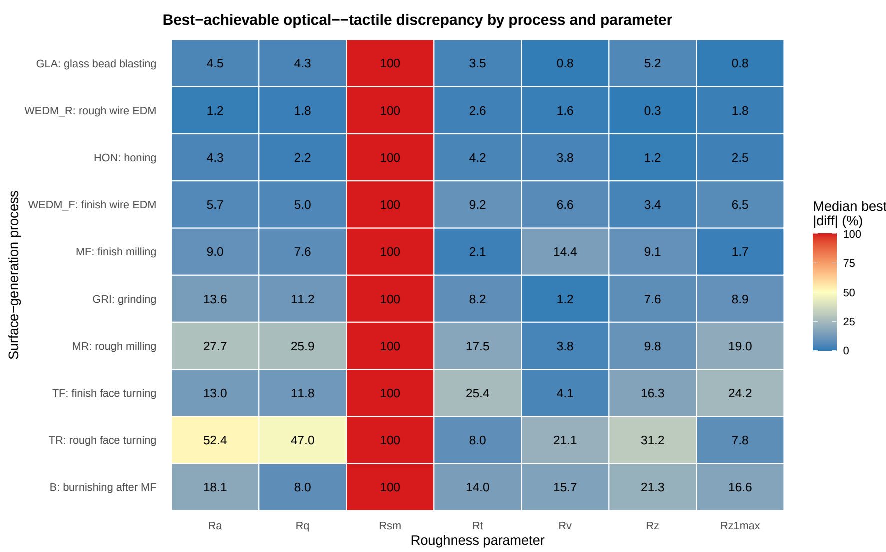

Figure 1: Global process–parameter map of best-achievable optical–tactile median discrepancy for the seven strictly positive roughness parameters. Values are median $|\text{diff}|(\%)$ after selecting the best available system–illumination pair within each material–process–parameter group; colour scale is capped at 100% for readability.

Across the full dataset, interferometry-based measurements (*int*) appear most frequently as the best-achievable configuration for most parameters, and red/green illumination bands are most often associated with the best-achievable median $|\text{diff}|(\%)$ (Table 1). Process- and material-specific exceptions are also present, with other systems, notably focus variation (FV), becoming the most frequent best-achievable choice in selected strata (Tables 4–5). These frequencies describe patterns within the tested archive.

Figure 2 illustrates the main effects of the measurement system on the absolute percentage error using the representative case of R_a on Ti6Al4V after finish milling. Analogous plots for other material–process combinations are included in the supplementary materials.

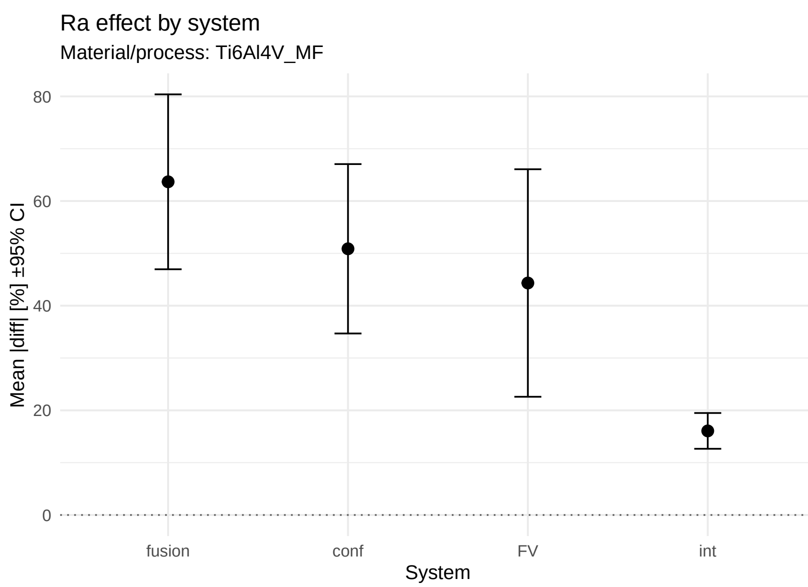

Figure 2: Main effects of the measurement system on absolute roughness error $|\text{diff}|(\%)$ for R_a on Ti6Al4V after finish milling

3.2 Representative example: Ti6Al4V after finish milling

To illustrate the system-to-system variability observed across the full dataset, a detailed example is shown for Ti6Al4V after finish milling, which represents a technologically relevant, moderately smooth surface. In the decision summaries, roughly half of all available R_a material–process groups reach $\leq 10\%$ disagreement after selecting the best optical system and colour (Table 1), while Ti6Al4V is among the more challenging materials in aggregate (Table 5).

For Ti6Al4V after finish milling, the spread of $|\mathrm{diff}|(\%)$ between systems is particularly clear. Some optical instruments yield errors close to the tactile reference under suitable illumination, whereas other configurations deviate markedly. Optical–tactile agreement is reported at the *system+illumination* level.

Figure 3 compares the distributions of $|\mathrm{diff}|(\%)$ for all optical systems against the tactile reference for Ti6Al4V after finish milling, emphasising the contrast between well-performing and outlying configurations.

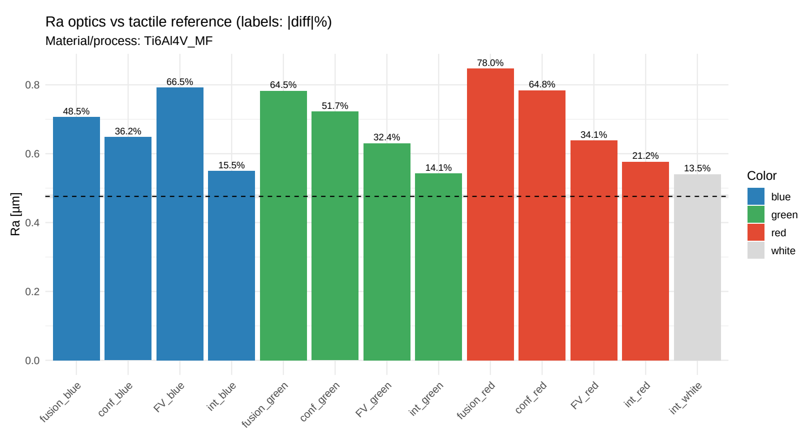

Figure 3: Comparison of absolute roughness error $|\mathrm{diff}|(\%)$ for R_a on Ti6Al4V after finish milling obtained from optical systems and the tactile reference

3.3 Influence of illumination wavelength

Across materials and processes, illumination colour can shift $|\mathrm{diff}|(\%)$ and, in a minority of cases, induces systematic trends with wavelength. In the full set of system-wise regressions across the analysed material–process–parameter groups, statistically significant slopes β_1 of $|\mathrm{diff}|(\%)$ versus wavelength were observed in 8.9% of fits (203/2290), indicating wavelength optimisation as an application-specific tuning step rather than a universal correction. The regressions span many parameter–surface groups and typically rely on only three or four colour levels per fit; individual significant slopes serve as screening evidence. The observed rate is only modestly above the nominal 5% false-positive rate, so the archive supports selective wavelength effects rather than a broad wavelength law.

Consistent with this, ANOVA effect sizes indicate that the measurement system explains substantially more variability in $|\mathrm{diff}|(\%)$ than illumination colour for most groups (median $\eta^2_{\text{system}} = 0.651$ vs. median $\eta^2_{\text{color}} = 0.106$ across all ANOVA groups, $n = 590$; Supplementary Table S1). Across the same 590 parameter–surface groups, the system effect size exceeds the colour effect size in 497 cases, whereas colour exceeds system in 93. Nevertheless, colour still matters operationally because the best-achievable configuration for many parameters is tied to a specific illumination band (Table 1), and these preferences can differ across process and material (Tables 4–5).

The full per-group ANOVA outputs (including degrees of freedom, F, p, and η^2) are reported in Supplementary Table S1, and the full set of system-wise wavelength-regression slopes is reported in Supplementary Table S2. Correlation coefficients were used as a complementary descriptive diagnostic alongside regression; to avoid redundancy, the paper reports regression slopes as the primary summary. Additional system- and colour-wise effect plots supporting follow-up comparisons are provided in the supplementary materials.

Figure 4 shows an example of the dependence of $|\mathrm{diff}|(\%)$ on illumination wavelength for R_a on Ti6Al4V after finish milling, together with fitted screening lines that illustrate the estimated trend for each optical system.

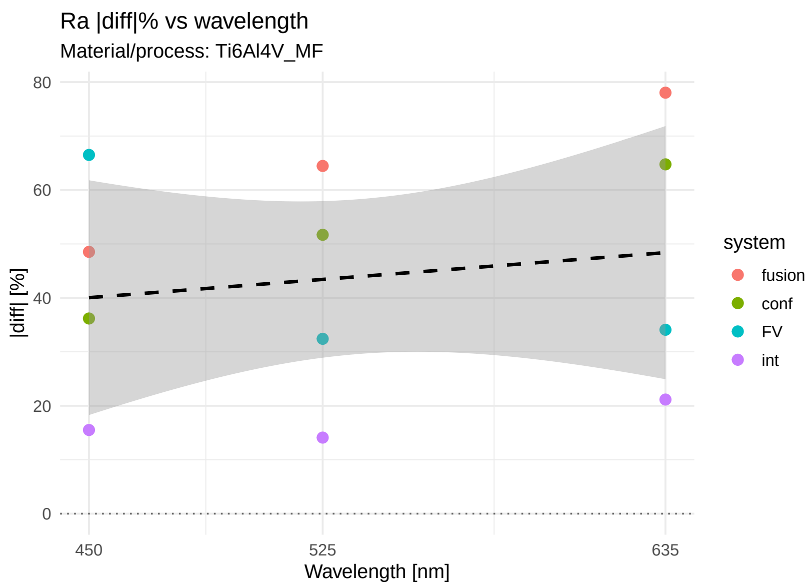

Figure 4: Dependence of absolute roughness error $|\text{diff}|(\%)$ for R_a on Ti6Al4V after finish milling on illumination wavelength for individual optical systems; lines indicate fitted screening trends

3.4 Technical distance between systems

The technical distance matrix derived from mean $|\text{diff}|(\%)$ highlights clusters of optical systems that behave similarly across the investigated materials and surface treatments. Systems within the same cluster tend to exhibit comparable error levels and wavelength trends, whereas isolated systems form distinct groups at large distances from the rest. Small distances indicate similar disagreement patterns within the analysed dataset, separately from matched metrological behaviour, shared traceability, or a common calibration strategy. As a representative example, Figure 5 shows the technical distance matrix for R_a on Ti6Al4V after finish milling. Equivalent matrices for other material–process combinations and parameters are available in the supplementary materials, and they provide a compact complement to the decision tables (Tables 1–5) by showing which systems behave similarly *within a given material–process context*.

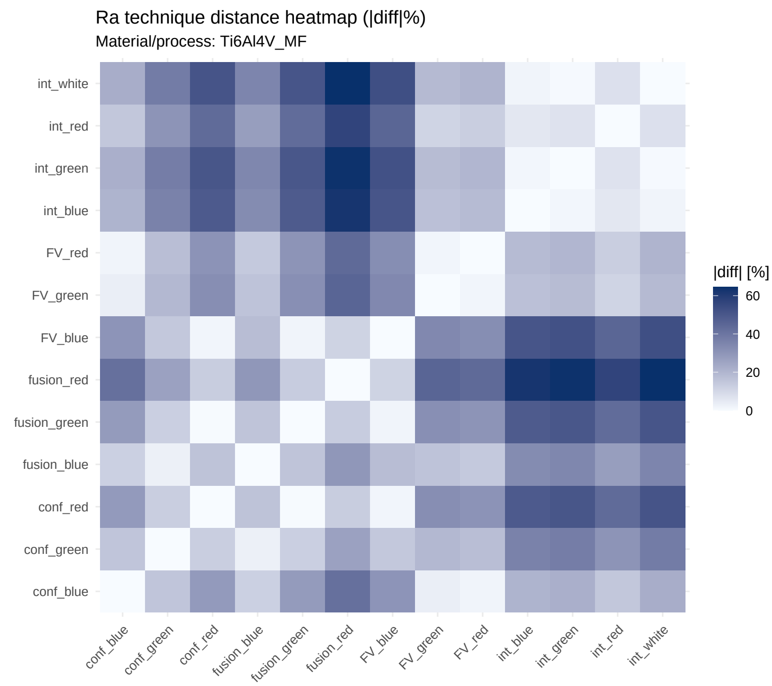

Figure 5: Technical distance matrix between optical and tactile systems for R_a on Ti6Al4V after finish milling, based on mean absolute roughness error $|\text{diff}|(\%)$

4 Discussion

A multi-material retrospective benchmark of optical workflows against an internal tactile baseline has been presented. Within the archive and its parameter-label comparison frame, three patterns are robust. First, system choice dominates the discrepancy structure across the archive. Second, illumination band matters, but mainly as a secondary, system-specific tuning variable. Third, performance is strongly parameter-dependent: amplitude-type parameters can enter a lower-discrepancy regime under selected configurations, whereas R_{sm} remains difficult.

The retained exported-value behaviour of R_{sm} is especially informative. It remains near the high-discrepancy limit even after allowing retrospective workflow selection, and the diagnostic segmentation shows the same pattern across optical-system families, materials, and processes (Table 2). The unit-scale sensitivity shows that this pattern contains a scale-harmonisation component: after multiplying retained optical R_{sm} values by 1000, the best-workflow median discrepancy decreases to 17.3%, but 15 of 59 material–process groups still remain above 30% discrepancy (Table 3). The manuscript therefore reports R_{sm} as a harmonisation-sensitive spacing result rather than as a corrected equivalence result. Spacing parameters are expected to be more sensitive than amplitude parameters to lateral sampling, profile-feature recognition, filtering, and the effective contact or optical footprint. The present archive identifies R_{sm} as a priority parameter for a future raw-profile and resolution-controlled validation study.

The decision tables identify the best-performing workflow observed within each parameter–surface group and serve as screening tools for follow-up validation. The ANOVA decompositions and wavelength regressions summarise patterns in an unbalanced archive, especially where only three or four illumination levels were available, and provide archive-level screening rather than causal or transferable correction models. Percentage-based R_{sk} summaries are retained in the archive-derived tables as secondary descriptors, separately from the strictly positive parameters.

The fixed-configuration sensitivity analysis separates archive screening from preselected-workflow use. In the screening view, the retrospective best-achievable summaries identify promising follow-up candidates. In the fixed-workflow view, one optical setup is evaluated before measurement across all available groups. The gap between these views is substantial: the best complete-coverage fixed workflow has a median discrepancy of 24.8%, whereas several amplitude parameters achieve best-achievable medians below 10% only after the workflow is allowed to vary by parameter–surface group. For a laboratory that must preselect one optical workflow, discrepancies of this order remain expected unless a narrower material–process range is validated prospectively. Optical workflow selection is therefore part of the metrological decision, and the manuscript decision bands are most appropriately read as validation priorities rather than as direct specification limits.

From a practical perspective, the archive supports workflow triage rather than workflow substitution. The results identify where optical configurations merit prospective testing, where system choice is decisive, and where tactile confirmation remains prudent. For laboratories using multiple instruments, the technical distance maps provide an operational summary of which workflows behave similarly within a given material–process context, separately from shared calibration or metrological equivalence.

A prospective study designed to support stronger tactile–optical equivalence claims would require a narrower and more rigorous protocol than the one available here. At minimum, such a study would need matched specimen-level raw profiles, harmonised filtering and form-removal settings, within-instrument and between-instrument repeatability, and a combined uncertainty model covering both tactile and optical workflows. System and illumination settings would also need to be fixed a priori, or selected in a separate development phase, rather than retrospectively optimised on the reporting dataset.

4.1 Interpretive scope

Several features define the scope of the present conclusions. The study compares exported roughness parameters in a harmonised comparison space defined by matched material–process identifiers and same-named parameter labels. A fully harmonised raw-profile measurand definition is incomplete across instrument pairs, and the reported discrepancies describe workflow-level disagreement. The tactile mean is an internal baseline derived from $n = 20$ repeats with an available repeatability descriptor (s_S, u_S). The tactile repeatability analysis indicates that random repeatability is usually smaller than the observed optical–tactile disagreements, while unmodelled tactile bias or calibration effects may still shift individual discrepancies. Optical repeat measurements were retained less consistently than tactile repeats, so the optical side of the benchmark is an archived-output comparison. The best-achievable decision tables are retrospective, with the best system+colour selected after observing the data; the fixed-configuration sensitivity analysis quantifies this selection effect. Bootstrap intervals in Supplementary Table S5 describe stability of the archive-level median summaries, not formal uncertainty of future optical use. Wavelength regressions and the descriptive summaries in Supplementary Tables S2–S3 are often based on only three or four colour levels per system and provide screening trends. Raw profiles, common filtering settings, complete optical instrument metadata, detailed process settings, and a combined tactile–optical uncertainty model are incomplete across the archive. Percentage-based R_{sk} summaries are intrinsically unstable near zero and are retained as secondary screens.

5 Conclusions

Within these limits, the main findings of the multi-material benchmark can be summarised as follows:

- **System choice is the primary driver of disagreement.** Across all analysed parameter–surface combinations, the ANOVA effect size for system is large (median $\eta^2 = 0.651$, interquartile range 0.357–0.831), and across the

590 parameter–surface groups with both system and colour effect sizes available, system has the larger η^2 in 497 cases (Supplementary Table S1).

- **Illumination colour is secondary but non-negligible.** The ANOVA effect size for colour is much smaller (median $\eta^2 = 0.106$, interquartile range 0.048–0.210), and colour has the larger η^2 in 93 of the same 590 groups; wavelength/colour optimisation is a system-specific tuning step rather than a universal correction.

- **Wavelength trends are present in a minority of cases.** Across all fitted system-wise regressions, statistically significant slopes of $|\text{diff}|(\%)$ versus wavelength are observed in 8.9% of cases (203/2290) at a nominal $\alpha = 0.05$. Rates are lowest for R_a (6.6%) and higher for amplitude parameters such as R_t (12.2%), R_z (11.4%) and R_{z1max} (10.9%) (Supplementary Table S2); these fits are screening summaries based mainly on three or four wavelength levels.

- **Parameter dependence is substantial.** Best-achievable discrepancy differs sharply across the strictly positive parameters: amplitude-type measures such as R_t, R_v, R_z, and R_{z1max} achieve median best discrepancy of 4.8–6.6%, R_a and R_q remain below 10% in median terms, and retained exported-value R_{sm} remains at 99.9% (Table 1). A post hoc optical R_{sm} x1000 sensitivity reduces the best-workflow median to 17.3%, while 25% of material–process groups remain above 30% (Table 3). Percentage-based R_{sk} summaries are retained for archive completeness as secondary descriptors, separately from the strictly positive parameters. Parameter-specific conclusions are preferred over generalisation from a single roughness parameter.

- **Retrospective selection materially improves the apparent result.** In the fixed-configuration sensitivity analysis, the best single workflow with complete coverage is interferometry with green illumination, but its median discrepancy is still 24.8% across the 413 strictly positive parameter–surface groups (Table 6). The decision tables prioritise candidate workflows for prospective validation rather than one universal optical setup.

- **The archive supports a limited internal reference-uncertainty treatment.** Across all 472 parameter–surface groups in the manuscript subset, the tactile reference is based on $n = 20$ repeats. For the seven strictly positive parameters, the relative standard error of the tactile mean has median 1.45% (Q1–Q3 0.85–2.44%), indicating that tactile repeatability is typically smaller than the observed optical disagreements.

- **The main output is a benchmark for follow-up selection.** The decision summaries capture the lowest disagreement observed within the tested system+colour combinations for each parameter–surface group and prioritise candidate optical workflows for prospective validation. Fixed-setup performance and tactile/optical equivalence remain prospective-validation questions.

Overall, the study identifies where lower optical–tactile discrepancy was observed within the tested configurations and where tactile confirmation remained preferred. Its main contribution is a practical benchmark for selecting candidate optical workflows and prioritising follow-up validation.

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Supplementary materials

The following supporting information is available online at the journal website:

- **Supplementary S1**: additional tables and figures extending the Results. The material includes (i) two-way ANOVA main effects and η^2 effect sizes for the factors system and colour (Supplementary Table S1), (ii) system-wise slopes of $|\text{diff}|(\%)$ versus illumination wavelength (Supplementary Table S2), (iii) descriptive statistics by system (Supplementary Table S3), (iv) parameter-wise internal tactile repeatability summaries (Supplementary Table S4), and (v) bootstrap percentile intervals for aggregate median best-achievable discrepancies (Supplementary Table S5), together with representative plots for selected material–process combinations. The supplementary tables are generated from the same archived output files as the manuscript tables, and the generation script is included in the public repository.

Conflicts of interest

The authors declare no conflict of interest.

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Data availability

The R/Python analysis scripts, configuration files, manuscript and supplement sources, generated CSV outputs, generated tables, and generated figures supporting the findings are provided in a public GitHub repository. The archived versioned release is available through Zenodo at DOI 10.5281/zenodo.19844491. Raw `.sur` surface files are excluded from the public repository as original instrument exports with size and transfer constraints; the derived CSV outputs needed to reproduce the manuscript tables, fixed-workflow sensitivity analysis, global heatmap, and supplementary tables are included. The excluded raw `.sur` files are available from the authors on reasonable request, subject to data-transfer constraints.

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